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Unit 3 - Designing Solutions

PDF Documents

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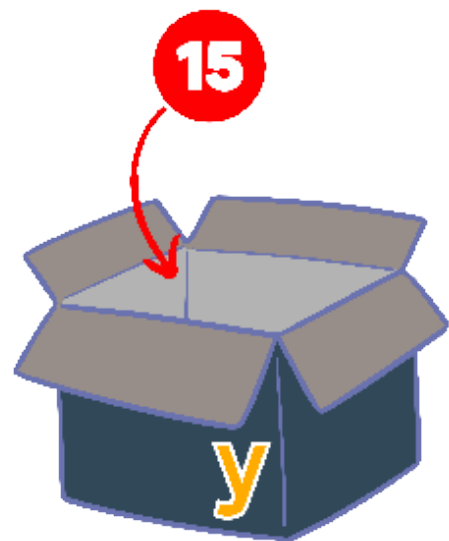
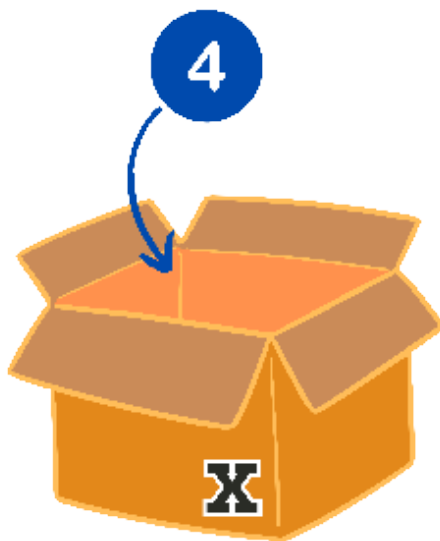


UNIT 3 DESIGNING SOLUTIONS

● Programming Concepts

In mathematics, variables such as **x** and **y** are used to represent unknown values that can change. For instance, in the equation $x + 2 = 5$, **x** is a placeholder for a value we need to discover. Once we solve the equation, we find that **x** equals 3.

✓ VARIABLES



Similarly, in programming, a variable is a name assigned to a storage area in the computer's memory where data can be stored, modified, and retrieved later. Just like in math, the value of a programming variable can change as the program runs. For example, if we set a variable **x** in a program with $x = 5$, we can later change it to $x = 10$ based on some conditions or calculations performed during the execution of the program.

EXPLANATION WITH CODE EXAMPLE

INCREMENT AND DECREMENT OPERATORS

Increment and decrement operators are used to adjust a variable's value by a fixed amount, typically 1, which simplifies coding in repetitive tasks. These operators are especially useful in loops for iterating over sequences.



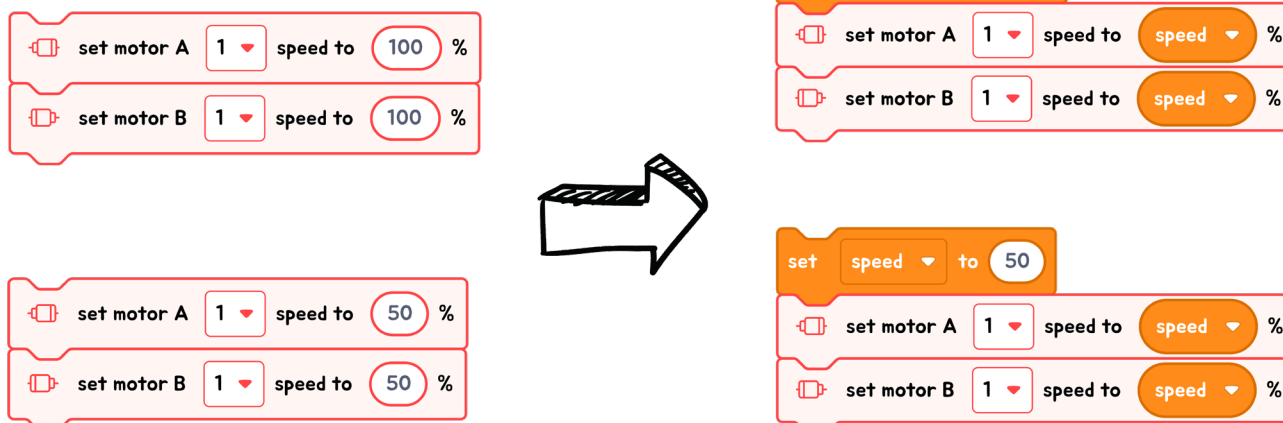
- The block on the left, known as the increment operator, increases the variable's value by 1.
- The block on the right, known as the decrement operator, decreases the variable's value by 1.



HOW ARE VARIABLES USED?

Variables make it easier to update and manage your code. Instead of repeatedly changing the same piece of information everywhere it appears in your program, you can use a variable. This means you only need to update the value in one place, and every part of your program that uses this variable will automatically use the new value.

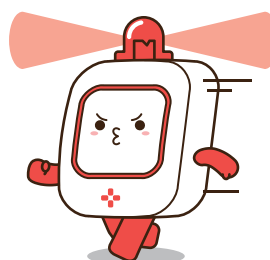
EXAMPLE:



Imagine you have a video game where a car is traveling at a speed of 100 units. If you want to slow the car down to 50 units, using a variable for speed is very helpful. Instead of finding every place in your code where the speed is set to 100 and changing it to 50, you simply change the variable 'speed' from 100 to 50 once. Now, every part of your game that uses this 'speed' variable will automatically update to the new speed.

THINK ABOUT IT!

Imagine needing to change the speed in a game code in many places—maybe 10 or even 100 times. Without variables, you'd have to update each one by hand, which is slow and can lead to mistakes. Using variables, you just change the speed once, and it updates everywhere automatically. This saves time and reduces errors, making it easier to manage your code as it gets more complex.





TIPS

The flowchart from the previous page continues on this page.

